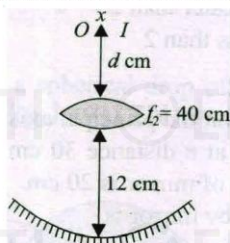
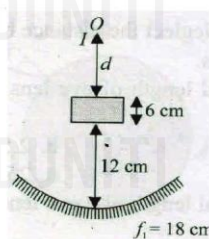


Paragraph Q17 to Q19

A convex lens of focal length 40 cm is held at a distance 12 cm coaxially above a concave mirror of focal length 18 cm

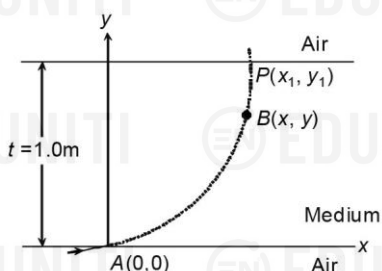


17. A luminous point object placed d cm above the lens on its axis given rise to an image coincident with itself, then the value of d is equal to
 (a) 15 cm (b) 30 cm
 (c) 24 cm (d) 18 cm
18. If the convex lens is replaced by a concave lens of focal length 60 cm, it gives rise to an image coincident with itself. Then the value of d is
 (a) 20 cm (b) 30 cm
 (c) 40 cm (d) 60 cm
19. If the convex lens is replaced by a glass plate of thickness 6 cm, refractive index $\mu = \frac{3}{2}$ and gives rise to an image coincident with itself, then the value of d is equal to
 (a) 20 cm (b) 30 cm
 (c) 40 cm (d) 60 cm



Paragraph Q20 to Q22

A ray of light travelling in air is incident at incident angle 90° on a long rectangular slab of a transparent medium of thickness $t = 1.0$ m as shown in the figure. The point of incidence is the origin $A(0, 0)$. The medium has a variable index of refraction $n(y)$ given by, $n(y) = [ky^{3/2} + 1]^{1/2}$, where $k = 1.0$ (m) $^{-3/2}$. The refractive index of air is 1.0.



20. The relation between θ (if slope of the trajectory of the path of ray at a point $B(x, y)$ in the medium is $\tan \theta$) and refractive index n at point B is

$$\begin{aligned} \text{(a)} \quad \frac{\sin \theta}{\theta} &= \frac{\sqrt{n^2 - 1}}{\cos^{-1}(1/n)} & \text{(b)} \quad \frac{\cos \theta}{\theta} &= \frac{\sqrt{n^2 - 1}}{\cos^{-1}(1/n)} \\ \text{(c)} \quad \frac{\tan \theta}{\theta} &= \frac{\sqrt{n^2 - 1}}{\cos^{-1}(1/n)} & \text{(d)} \quad \frac{\cot \theta}{\theta} &= \frac{\sqrt{n^2 - 1}}{\cos^{-1}(1/n)} \end{aligned}$$

21. The equation of the trajectory of the ray in the medium is
 (a) $4y^{1/4} = k^{1/2}x$ (b) $4y^{1/2} = k^{1/2}x$
 (c) $4y^{1/4} = k^{1/2}x^{1/2}$ (d) $4y^{1/2} = k^{1/2}x^{1/2}$
22. After coming out of medium, ray of light will move
 (a) parallel to x-axis
 (b) parallel to y-axis
 (c) at an angle 45° with x-axis
 (d) on a zig-zag path

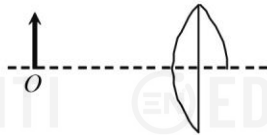
More than one correct

23. The distance between a screen and an object is 120 cm. A convex lens is placed closed to the object and is moved along the line joining object and screen, towards the screen. Two sharp images of the object are found on the screen. The ratio of magnification of two real images is 1 : 9. Then,
 (a) focal length of the lens is 22.5 cm
 (b) smaller image is brighter than the larger one
 (c) larger image is brighter than the smaller one
 (d) brightness of both the images is same
24. The concave and convex surfaces of a thin concave-convex lens of refractive index 1.5 has radius of curvature 50 and 10 cm respectively. The concave side is silvered and placed on a horizontal surface as shown



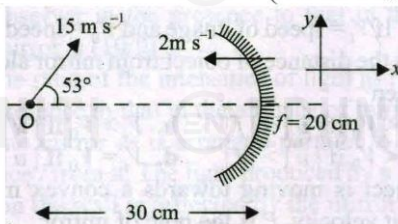
- (a) focal length of lens is 25 cm
 - (b) focal length of the combination after silvering is 25 cm
 - (c) image of the object at 50 cm will coincide with itself
 - (d) image will not coincide in this type of silvered lenses
25. A convex lens made of glass ($\mu_g = 3/2$) has focal length f in air. The image of an object placed in front of it is inverted real and magnified. Now the whole arrangement is immersed in water ($\mu_w = 4/3$) without changing the distance between object and lens. Then
 (a) the new focal length will become $4f$
 (b) the new focal length will become $f/4$
 (c) new image will be virtual and magnified
 (d) new image will be real, inverted and smaller in size

26. One fourth part of an equiconvex lens of focal length 100 cm is removed as shown in the figure. An object of height 1 cm is placed in front of the lens. It is observed that all the images are of equal height. Then



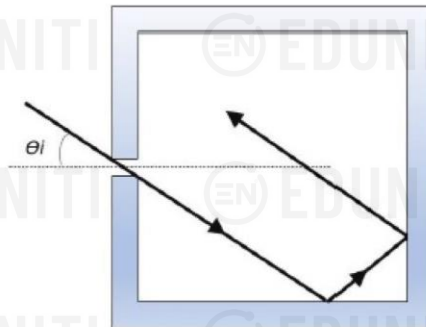
- (a) Object is at a distance of $400/3$ cm from the lens.
- (b) The magnitude of magnification produced by upper and lower part is equal.
- (c) The number of images formed is two.
- (d) The product of magnification of both the lenses is negative.

27. In the situation as shown in Fig. $\left(\cos 53^\circ = \frac{3}{5}\right)$



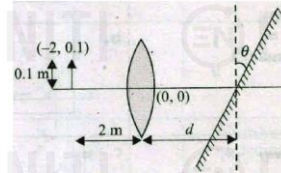
- (a) velocity of image wrt mirror is $-22\hat{i} - 24\hat{j}$
- (b) velocity of image wrt mirror is $-44\hat{i} - 24\hat{j}$
- (c) velocity of image wrt ground is $-46\hat{i} - 24\hat{j}$
- (d) velocity of image wrt ground is $-24\hat{i} - 24\hat{j}$

29. A monochromatic beam of light enters a square enclosure with mirrored interior surfaces at an angle of incidence θ_i ($\neq 0$) (see the figure below). For some value(s) of θ_i , the beam is reflected by every mirrored wall (other than the one with the opening) exactly once and exits the enclosure through the same hole. Which of the following statements about this beam is correct?



- A) The beam will not come out of the enclosure for any value of θ_i .
- B) The beam will come out for more than two values of θ_i .
- C) The beam will come out only at $\theta_i = 45^\circ$.
- D) The beam will come out for exactly two values of θ_i .

28. A convex lens of focal length 1.5 m is placed in a system of coordinate axis such that its optical centre is at origin and principal axis coinciding with the x-axis. An object and a plane mirror are arranged on the principal axis as shown in Fig. Find value of d (in m) so that y-coordinate of image (after refraction and reflection) is 0.3 m. (Take $\tan \theta = 0.3$)



Answer Key

- 17. a
- 18. c
- 19. a
- 20. c
- 21. a
- 22. a
- 23. ab
- 24. abc
- 25. ac
- 26. abcd
- 27. bc
- 28. 5
- 29. d